



ISTE 436 – Database Management and Access

Spring 2026

Project Description

Due date: April 29, 2026 at 12:00 PM

Mission

In this project, you will build and administer an enterprise-level database system. You will apply all the database concepts cultivated from your lab exercises and demonstrate a fully-functional relational database management system.

Statement

Define a simple business scenario that typically uses relational databases for business basic operations. Consult with your instructor and proceed only after the instructor approves your idea. Then, model the database using an ER-Diagram. After that, use AWS EC2 to create and launch an Oracle database. Finally, demonstrate different operations. Below are detailed steps.

1. (10 points) Define a business scenario where relational databases are needed.
 - a. Describe the scenario in two or three paragraphs.
 - b. Define a minimum of 4 entities along with their attributes.
 - c. Clearly define attribute types. You must have the following types somewhere:
 - i. Primary/foreign keys
 - ii. Integer, decimal, date
 - iii. Text of limited choices (e.g., Fall, Winter, Spring, and Summer)
 - iv. Text of up to 20 characters long (e.g., names)
 - v. Text of up to 255 characters long (e.g., comments)
 - vi. A single character of limited choices (e.g., Y/N, T/F, or L/M/S).
 - d. Use a computer-aided tool to draw an ER-Diagram that shows all the entities, their attributes, and the relationships and their cardinalities (1:1, 1:M, M:M). Use the Crow Foot notation.
2. (10 points) Ideally, if you were to distribute datafiles across multiple disks based on use, how would you go about this activity. Complete the weight assignment activity by filling Tables 1-2 and 2-1 from document "ExerciseW0501-Wrights.docx". Note that this activity will not impact your implementation.
3. (10 points) Use a new AWS instance for this project. Do not follow on the same instance that you have been using in the labs. You may look for an "Oracle Linux" instance, or you may launch the original clean instance you started with in this course to make use of its stored scripts. Choose a "Medium" instance. Assign a new AWS S3 volume to access the instance.
4. (10 points) Use DBCA to create your Oracle database. Create at least four user roles:
 - a. An admin who has the highest level of power.
 - b. A developer who is capable of creating tables.
 - c. Two end users who interact with the database to insert or retrieve data.
5. (10 points) Use the developer role to create tablespaces and tables. Fill in some tables with dummy data from a text file or a CSV file. Do not use INSERT INTO for bulk data insertion.

6. (10 points) Configure the database to listen for remote connections. Simultaneously connect two end users to the database. Make sure the end users are authenticated and authorized to do their relevant tasks. Explain the importance of such a security practice.
7. (20 points) Manage and demonstrate concurrent transactions.
8. (10 points) Demonstrate the following SQL queries:
 - a. Perform a meaningful join operation.
 - b. Showcase the use of grouping, sorting, displaying unique records, two numerical functions, and two string manipulation functions. You do not have to create a single query for each.
 - c. Create a view of your own choosing.
9. (10 points) Backup:
 - a. Explain the importance of backup.
 - b. Develop a backup strategy.
 - c. Backup your database.
 - d. Explain the rationale behind your choice of your backup strategy.

Deliverable

This is a group-based project. Form a group of three students, and all of you must join an available group on myCourses → Groups. Your team will submit a PDF report, PPT presentation, and all supporting files, such as SQL and Linux script files.

Report

Your report will be marked out of 100 points, which constitutes 10% of the total course grade. Your report must contain the following components/sections.

1. Include a cover page/slide with course title, assessment title, students' names and IDs, date, and group#.
2. Page 2: Manifesto
 - a. Clearly state the contributions of each team members.
 - b. If all team members contributed equally to the project, specify how.
 - c. If (b) above is not the case, then the manifesto must specify who did what. Students will be graded based on their contribution to the team.
 - d. A complete work is still expected regardless who contributed to the work.
 - e. Not submitting a manifesto results in losing 20% of the project grade.
3. Page 3: Table of contents that clearly list each section/subsection and its related page.
4. Page 4: Create a section for each step. Each step and choice must be accompanied by proper explanation and screenshots.
5. Your report must be clearly written and organized, otherwise, marks will be deducted.

Presentation

Along with your report, you must also submit a presentation. All team members must be present in-person during the presentation. Failure to do so will result in a **zero on the whole** project for the missing team member. Your team will present their work to me as per the specified deadline on the cover page of this document. The work must be submitted on time regardless when the scheduled presentation timeslot is.

Your presentation will be marked out of 100 points, which constitutes 15% of the total course grade. The presentation will be evaluated based on the following criteria:

1. (20 points) The presentation must be organized:
 - a. Cover slide that has course#, students' names, group#, and date
 - b. Manifesto
 - c. Table of contents
 - d. Summary of business scenario
2. (60 points) Speak clearly, no stuttering, muttering, or reading from notes
 - a. Your presentation must reflect your full understanding of intricacies involved in building and administering the database system.
 - b. You must be able to demo all the basic business operations to your scenario.
 - c. You must be able to answer the instructor's questions when asked.
 - d. Inability to answer instructor's questions will negatively affect your presentation grade.
 - e. You are allocated 10 minutes + 5 minutes of Q&A. Be strict with timing.
3. (20 points) The presentation must include snippets of each major script written.

Submission

This is a group-based project. You will work **in a team of exactly three students**. Your team will submit (1) a PDF report along with all supporting files, such as SQL and Linux script files, and (2) PPT presentation, to (1) myCourses → Assignments → Report, and (2) myCourses → Assignments → Presentation

Note that only a single submission per team is required. You will submit your work as a group, not individually. The submission of all files must be done **by the deadline date and time** shown on the cover page of this document. The file name structures are as follows, where “Group#” is your group number: Group#_Report.pdf, Group#_Presentation.ppt, Group#_SuppFile#.

Turnitin will be active, and you will be able to see your Similarity Score. Similarity Scores above 20% will result in an automatic mark of zero. You have the option to resubmit your report if you wish to modify it before the deadline. The portion of the Similarity Score that generates from copying parts of this report or listing references, does not contribute toward the 20% limit. Moreover, you are **not allowed** to use AI tools to draw or write any component of your report.

Summary of Grading

Your work will be marked as per the below distribution:

Step	Points
Steps 1-3	10 each
Step 4, 7	20 each
Steps 5-6, 8	10 each
Total	100
Note: Not including a manifesto will result in deducting 20 points	

Table 1: Report Grading

Requirement	Points
Organization	20
Answering instructor's questions	60
Snippets from scrips	20
Total	100

Table 2: Presentation Grading

Rubric

Each step will be evaluated against the below rubric:

Rubric	R1	R2	R3	R4	R5
	- The step has not been attempted, or - The provided solution contains several major flaws that reflect critical weaknesses	The provided solution is abstract and requires major improvement or additional details	The provided solution suffers from one or two major flaws, but enough details are provided	The provided solution suffers from one or two minor flaws, and enough details are provided	The provided solution is complete, does not suffer from any flaws, and includes all required details
Points	0	40	60	80	100